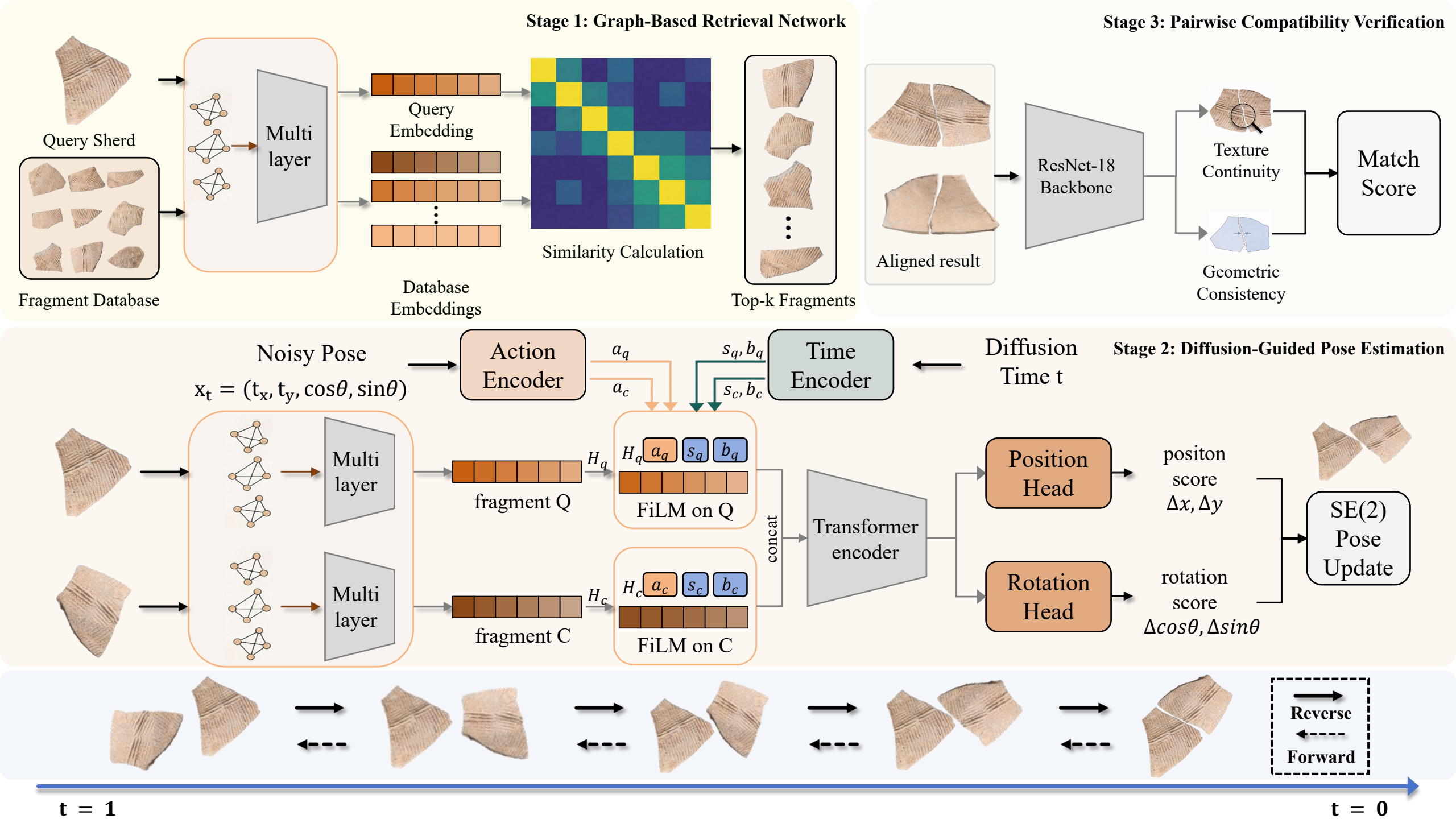
Stage 1: Graph-Based Retrieval Network
Query Sherd
Fragment Database
Multi layer
Query Embedding
Database Embeddings
Similarity Calculation
Top-k Fragments
Stage 3: Pairwise Compatibility Verification
Aligned result
ResNet-18 Backbone
Texture Continuity
Geometric Consistency
Match Score
Stage 2: Diffusion-Guided Pose Estimation
Noisy Pose
$x_t = (t_x, t_y, \cos\theta, \sin\theta)$
Action Encoder
Time Encoder
Diffusion Time t
a_q
a_c
s_q, b_q
s_c, b_c
Multi layer
Multi layer
fragment Q
fragment C
H_q
H_c
FiLM on Q
FiLM on C
concat
Transformer encoder
Position Head
Rotation Head
positon score $\Delta x, \Delta y$
rotation score $\Delta cos\theta, \Delta sin\theta$
SE(2) Pose Update
Reverse
Forward
t = 1
t = 0